

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Summer-2023 Final Examination
Program: B.Sc. in CSE (Batch – 15th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 0533 1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x1=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Knowledge

1. Which one is used as a proton in nuclear reaction?

A) 1_1H
B) 2_1H
C) 3_1H
D) 1_0p
E) 4_2He

Cognitive Analysis

2. When the velocity v of the body is negligibly small as compared to light velocity c , what will be relation between stationary length, L_0 and moving length, L ?

A) $L < L_0$
B) $L > L_0$
C) $L = L_0$
D) $L < L_0 < 0$
E) $L > L_0 > 0$

Cognitive understanding

3. Photo current increases with the increase of the _____.

A) stopping potential.
B) frequency of the incident photon.
C) wavelength of the incident photon.
D) intensity of the incident light.
E) work function of the metal.

Cognitive Applying

4. What will be the angle between the polarizer and the analyzer when a beam of natural light is reduced to 0.125 of its original intensity?

A) 90°
B) 69°
C) 45°
D) 30°
E) 0°

Cognitive Comprehension

5. What is the activity of 1 mg of iodine-131 (Half-life = 8.019days)?

A) $1.6 \times 10^{12} \text{Bq}$
B) $2.6 \times 10^{12} \text{Bq}$
C) $3.7 \times 10^{12} \text{Bq}$
D) $4.6 \times 10^{12} \text{Bq}$
E) $5.7 \times 10^{12} \text{Bq}$

Cognitive Applying

6. If N_0 the initial number of substances of half-life 2years, then the amount of substance left after 6 years is
- A) $N_0/2$
 - B) $N_0/4$
 - C) $N_0/8$
 - D) $N_0/10$
 - E) $N_0/12$

Cognitive understanding

7. The phase difference between two coherent sources should be _____ .
- A) zero
 - B) π
 - C) 2π
 - D) $\pi/2$
 - E) $3\pi/4$

Cognitive Knowledge

8. What is unit of grating constant?
- A) cm
 - B) cm^{-1}
 - C) J
 - D) Js^{-1}
 - E) N

Cognitive Knowledge

9. The central fringe of the double slit experiment must be _____.
- A) thin
 - B) thick
 - C) conical
 - D) dark
 - E) bright

Cognitive understanding

10. Which of the following can be considered as inertial frame reference?
- A) Freely falling body
 - B) Projectile
 - C) Earth
 - D) Spinning top
 - E) Pendulum in air

Signature of Examiner

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Summer – 2023 Final Examination
Program: B.Sc. in CSE (Batch – 14th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 1201

PART-B: SAQ

Time: 1 Hour and 50 Minutes

Total Marks: 5x6=30

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. a) Clarify stopping potential. [1]
b) Draw the kinetic energy of photo electron vs frequency of incident light curve in photo-electric effect. [2]
c) Work function of sodium and potassium are 5.3eV and 2.2eV respectively. UV light of 3.5×10^{-7} m wavelength incident on the both metal surfaces. Is it possible to happen the photo-electric effect from both metal? – Justify your answer. [3]
2. a) List the basic components of nuclear reactor. [1]
b) Illustrate the nuclear chain reaction and mention the controlling mechanism of it. [2]
c) Short form of a nuclear reaction: ${}^{10}_5B(\alpha, n){}^{13}_7N$; (Where, Mass of proton = 1.00728 amu, Mass of neutron = 1.00867 amu, Mass of the helium nucleus = 4.00276 amu)
i) Calculate the radius of the product nucleus. [1]
ii) Compare the binding energies of the incident and outgoing particles. [2]
3. a) State the postulates of special theory of relativity. [2]
b) Justify the statement that “the force is invariant in all inertial frame of reference”. [2]
c) Show by the theory of relativity that the velocity of a body cannot be more than the velocity of light. [2]
4. a) Identify the change of beta decay from ${}^{14}C$ and give an use of this transformation. [1]
b) Characterize the radioactive decay curve to explain term “the radioactivity never ends”. [2]
c) Calculate the time in which the activity of a sample of thorium reduces to 90% of its original value. Assume the half-life of thorium to be 1.4×10^{10} years. [3]
5. a) Formulate the pointing vector. [1]
b) Distinguish intensity distribution curves of interference and diffraction of light. [2]
c) Light from a sodium vapour lamp ($\lambda = 589$ nm) forms an interference pattern on a screen 0.8m from a pair of slits. The bright fringes in the pattern are 0.35 cm apart. What is the slit separation? [3]
6. a) Write a short note on “double refraction of light”. [2]
b) “For the polarizing angle of incidence, the reflected & refracted rays are 90° apart” – explain the statement in according with Brewster law. [2]
c) The refractive index for a plastic is 1.25. calculate the angle of refraction for a ray of light incident at polarizing angle. [2]